

optimised ruggedness in high power systems. The ICs feature high output current capability, making them suitable for high-power applications,



such as HVAC systems, heat pumps, servo drives, and industrial inverters.

The use of Silicon on Insulator technology ensures robustness and immunity to negative transient voltages. The ICs comply with RoHS standards, exhibit resistance to ESD and moisture, and offer reliable protection against voltage transients. The product line includes four versions in two packages, with different current drive capabilities. The devices have excellent switching performance, ample creepage and clearance distances, and incorporate features like bootstrap diodes, precise overcurrent protection, and active output voltage clamping. The Eice-DRIVER series will enable designers to come up with more robust design, reduced bill of material, and faster time to market.

Infineon

<https://www.infineon.com>

Synchronous rectifier power IC

The GaNSense Control family from Navitas is an integrated synchronous



rectifier power IC that combines next-gen gallium nitride (GaN) power with high-frequency control functionality. It offers the benefits of the monolithically-integrated GaN power FET and GaN drive, plus control and protection circuits in a single surface-

mount package for high-density charger, adaptor, and auxiliary power applications. It will enable designers to build compact fast chargers, with better heat management and longer product life. The IC works at a high frequency of 225kHz, which results in a compact system size and weight. Integrated features such as loss-less current sensing, high-voltage start-up, and elimination of VDD inductor reduce component count and increase system efficiency. The GaNSense control offers a transient voltage breakdown of up to 800V and no PCB hotspots.

Navitas Semiconductor

<https://navitassemi.com>

SiC MOSFETs and PIMs

Onsemi has introduced a new generation of 1,200V EliteSiC silicon carbide (SiC) M3S devices, including MOSFETs and power integrated modules (PIMs).



The components are designed to offer higher efficiency and lower cost while offering rapid switching speeds and the

industry's lowest drain-to-source resistance per switch position, making them ideal for applications such as 800V electric vehicle onboard chargers, EV charging infrastructure, solar power, and energy storage systems. The PIMs are specifically designed for industrial applications and provide high-power conversion stages with optimised direct bonded copper designs for balanced current sharing and thermal distribution. The MOSFETs utilise M3S technology for high-speed switching. By reducing application

footprint and cooling requirements, Onsemi's EliteSiC M3S products help designers achieve higher efficiency and increased power densities in their high-power converters.

Onsemi Semiconductor Mfg Co.

<https://www.onsemi.com>

GaN e-mode FETs

Nexperia has released a series of gallium nitride field effect transistors (GaN FETs) in enhancement mode



(e-mode) suitable for both low voltage (100-150V) and high voltage (up to 650V)

applications. These GaN FETs offer high power efficiency and compact size, making them ideal for power conversion applications and reducing the bill of material. These FETs will enable engineers to build compact and efficient DC-DC converters in data centres, faster charging for e-mobility and USB-C devices, compact lidar transceivers, lower-noise class D audio amplifiers, and power-dense consumer devices like smartphones, laptops, and gaming consoles. The 650V-rated e-mode FET in the series is suitable for high-voltage applications.

Nexperia

<https://www.nexperia.com>

Multilayer ceramic capacitors

Murata GRM series of multilayer ceramic capacitors (MLCCs) combine elevated capacitance levels with miniaturised form factors and enhanced temperature performance. These MLCCs offer capacitance values of 10µF in a 0201-inch/0603M package size, representing a 65% size reduc-



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